

Trucking Multi-Mode Routing (FTL / PTL / LTL)

MVP Formal Model — Phase 1, v1

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Draft — 2026-05-16

Abstract

Formal specification of the optimization model for the multi-mode trucking routing problem, covering three distinct service tiers: Full Truckload (FTL), Partial Truckload (PTL, also called Volume LTL), and Less-than-Truckload (LTL). The model is a Binary Multi-Commodity Flow problem on a commodity-specific subgraph, with FTL/PTL bin-packing-style consolidation, LTL per-shipment pre-computed cost reflecting density-based class and weight-break tiered pricing, and hard tender-refusal constraints. The model parallels Ocean FCL (`model/ocean_fcl_routing.tex`), Ocean LCL (`model/ocean_lcl_routing.tex`), and Air Freight (`model/air_freight_routing.tex`) for notational consistency, with significant additions reflecting the realities of US-domestic trucking: tender acceptance probability, contract vs. spot capacity, dock appointment / MABD windows, and carrier-lane service availability. The design was refined after an adversarial critique focused on modeling realism (§9).

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1 Problem Statement

Given a set of trucking shipment requests (commodities), each defined by origin location, destination location, volume, weight, piece dimensions, cargo type, and time constraints, find a minimum-expected-cost assignment of commodities to trucking arcs across three service modes such that:

- every commodity is delivered within its delivery appointment window (Must-Arrive-By-Date / MABD),
- each FTL or PTL truck respects volume and weight capacity,
- LTL hard refusal rules are honored (linear-foot caps, piece dimensions, weight ceiling),
- LTL pickup cutoffs and carrier-published transit times are respected,
- contract FTL lane allocations are not exceeded,
- carrier-lane service availability (residential, liftgate, limited access) is respected,
- shipper-specific FAK class overrides apply where contracted,
- expected cost includes tender acceptance probability and re-tender markup, and
- optional per-commodity budget caps are respected.

Three trucking modes — distinct pricing and operational regimes.

Mode	Description
FTL	Full Truckload. One shipper's freight (or one consolidated group) fills a hired truck; direct origin-to-destination move. Per-mile pricing. Typical $\geq 10,000$ kg or ≥ 26 m ³ .
PTL	Partial Truckload / Volume LTL. The 10,000–25,000 kg or 6–18 pallet band where freight is too large for traditional LTL but doesn't fill a truck. Direct move with no terminal handling; per-shipment pricing.
LTL	Less-than-Truckload. Carrier consolidates multiple shippers' freight through service-center network. Pickup, linehaul (possibly multi-hop within carrier network), delivery. Density-based class + weight-break tier pricing.

Forwarder vs. carrier perspective. This model is the freight forwarder's optimization, not the LTL carrier's. The forwarder *tenders* a shipment to a carrier; the carrier then routes it through their internal network. The forwarder does *not* decide which breakbulk hubs a carrier uses internally. Consequently, the LTL network here is modeled as a directed bipartite graph of (carrier, origin service center, destination service center) tuples with published transit times and rates — no intermediate hub nodes. This is a key correction to the textbook hub-routing approach (Powell & Sheffi 1983), which describes the *carrier's* load-planning problem.

MVP scope.

- US-domestic trucking. Cross-border (Mexico/Canada) deferred to P1.
- FTL: 53' dry van baseline; 48' as alternative equipment type.
- PTL: per-shipment pricing model (no equipment-type decision at MVP).
- LTL: density-based class (NMFC 2025 SDS) with FAK override; weight-break tier pricing with deficit-weight rule; minimum charge floor.
- Tender acceptance probability per (carrier, lane, day) as parameter; expected cost reported (see §4.8).
- Time discretization to days (LTL/FTL scheduling does not require hour-level precision).

- Container-to-chassis: one ocean container per chassis per truck (hard).
- Probabilistic transit time, driver Hours of Service, backhaul optimization, dedicated lanes, reefer/hazmat/oversize, pool distribution explicitly deferred.

2 Graph Structure

2.1 Physical Network Graph $G(N, A)$

$$N = N_O \cup N_{SC,O} \cup N_{SC,D} \cup N_D$$

N_O — **Origins** Shipper doors (or POL/POD/CFS upstream from another mode).

$N_{SC,O}$ — **LTL origin service centers**
Per-carrier origin service center serving the shipper's ZIP. One node per (carrier, origin SC) tuple.

$N_{SC,D}$ — **LTL destination service centers**
Per-carrier destination service center serving the consignee's ZIP.

N_D — **Destinations**
Consignee doors (or POL/POD/CFS downstream to another mode).

Remark 1 (No intermediate hub nodes). Modern freight-forwarder routing does not select breakbulk hubs inside a carrier's network. The carrier's internal hub-and-spoke is opaque to the forwarder; published transit times and rates already embed it. LTL arcs are therefore direct origin-SC $\rightarrow$ destination-SC edges, with carrier-published transit time. This is the principal departure from the Powell-Sheffi LTL load-planning literature (which addresses the carrier's internal problem).

$$A = A_{PU} \cup A_{LH-FTL} \cup A_{LH-PTL} \cup A_{LH-LTL} \cup A_{FD}$$

A_{PU} — **Pickup** Origin door $\rightarrow$ LTL origin SC (truck pickup leg of LTL service). For FTL/PTL, pickup is bundled into the linehaul arc (no separate pickup node).

A_{LH-FTL} — **FTL linehaul**
Direct origin region $\rightarrow$ destination region. One arc per (carrier, lane, equipment, contract/spot type).

A_{LH-PTL} — **PTL linehaul**
Direct origin region $\rightarrow$ destination region. Per-shipment pricing, no terminal handling. One arc per (carrier, lane).

A_{LH-LTL} — **LTL linehaul**
(SC_O, SC_D) pairs within a carrier's network. Carrier-published transit time.

A_{FD} — **Final delivery**
LTL destination SC $\rightarrow$ consignee door. For FTL/PTL, delivery is bundled into the linehaul arc.

2.2 Commodity-Specific Subgraph $G(N_k, A_k)$

Same construction principle as other modes. Arcs in A_k must be time-feasible, satisfy LTL hard refusal rules (P.6, P.7), and pass carrier-lane service-availability checks (P.12).

3 Sets and Indices

Symbol	Description
K	Set of shipments (commodities), indexed by k
N, A	All nodes and arcs
$A^k \subseteq A$	Feasible arcs for shipment k
$A_M^k \triangleq A^k \cap A_M$	Feasible arcs of mode $M \in \{\text{PU, LH-FTL, LH-PTL, LH-LTL, FD}\}$
$\mathcal{M}$	Trucking modes = $\{\text{FTL, PTL, LTL}\}$
$\mathcal{C}$	Set of trucking carriers (FedEx Freight, Old Dominion, XPO, Saia, Estes, R+L, ABF, Knight-Swift, etc.)
$\mathcal{E}$	Equipment types = $\{53', 48' \text{ dry van}\}$ (FTL/PTL); LTL uses carrier-internal equipment, opaque to model
$\mathcal{T}$	Time periods indexed by τ . MVP: day buckets ($\tau \in \{0, 1, 2, \dots, T_{\max}\}$, days).
Z	Set of contract FTL lane allocations $z \in Z$; each has (carrier, lane, period) tuple
N_k	Nodes incident to A^k

4 Parameters

4.1 Commodity Parameters

Parameter	Symbol	Description	Unit
Volume	v_k	Cargo volume	m ³
Weight	w_k	Gross weight	kg
Density	δ_k	w_k/v_k (computed)	kg/m ³
Pallet count	p_k	Standard pallets (48×40 in)	—
Linear feet	ℓ_k	Linear feet of trailer occupied if loaded LTL	ft
Max piece length	$L_k^{\max}$	Longest single piece dimension	ft
Max piece weight	$W_k^{\max}$	Heaviest single piece (HU) weight	kg
Cargo ready	t_k^{rdy}	Earliest pickup date	day index τ
Delivery window start	T_k^{open}	Earliest acceptable delivery (MABD lower bound)	day index τ
Delivery window end	T_k^{close}	Latest acceptable delivery (MABD upper bound / deadline)	day index τ
Origin	$o(k)$	$\in N_O$	—
Destination	$d(k)$	$\in N_D$	—
Cargo type	τ_k	General / DGR / temp / oversize / valuable	categorical
NMFC class	$\text{class}_k^{\text{NMFC}}$	Density-derived NMFC class (50–500) per 2025 SDS	—
FAK class override	$\text{class}_k^{\text{FAK}}$	Shipper-contract-collapsed class (if applicable)	—
Service flags	flags_k	Liftgate / inside / residential / limited access required	set
Budget cap	B_k	Hard cost ceiling; ∞ if not set	USD

Linear feet estimation.

$$\ell_k \triangleq \left\lceil \frac{p_k}{2} \right\rceil \cdot \frac{40 \text{ in}}{12 \text{ in/ft}} \approx 3.33 \cdot \lceil p_k/2 \rceil \text{ ft} \quad (1)$$

assuming pallets loaded two-wide in a 102'' trailer. Used in P.6 (linear-foot rule).

4.2 Equipment Parameters (FTL/PTL)

Equipment e	Internal volume V_e	Max payload W_e	Max linear ft L_e	Pallet capacity
53' dry van	96 m ³	19,500 kg	53 ft	26 std
48' dry van	84 m ³	19,500 kg	48 ft	24 std

Fill rate cap $\eta = 0.85$ for road consolidation (lower than ocean/air due to dunnage, pallet jack access lanes, and weight distribution constraints).

4.3 Pickup, Final Delivery (Ground) Arc Parameters

For each ground arc $(i, j) \in A_{\text{PU}} \cup A_{\text{FD}}$:

Parameter	Symbol	Description	Unit
Cost	c_{ij}^G	Per-shipment local trucking cost	USD
Transit	μ_{ij}^G	Days (typically 0–1)	days

4.4 FTL Linehaul Arc Parameters

For each $(i, j) \in A_{\text{LH-FTL}}$ on (carrier, lane, equipment, contract-type):

Parameter	Symbol	Description	Unit
Carrier	$\text{car}(i, j)$	Carrier ID	—
Equipment	$e(i, j)$	53' or 48' dry van	—
Contract type	$\text{ct}(i, j)$	Contract or spot	—
Distance	dist_{ij}	Lane distance	miles
Base rate	r_{ij}^{FTL}	Per-mile rate	USD/mi
Fuel surcharge	fsc_{ij}	Pct of linehaul, refreshed weekly	—
Transit time	μ_{ij}^{FTL}	Mean transit days	days
Tender accept prob	p_{ij}^{acc}	Probability tender accepted at quoted rate (§4.8)	—
Re-tender premium	ρ_{ij}^{re}	Multiplier on base if rejected (typ. 1.15–1.25)	—
Contract allocation	$\text{alloc}_{ij}(\tau)$	Trucks reserved on contract lane in week τ (if $\text{ct} = \text{contract}$)	trucks
Service flags supported	flagsAvail_{ij}	Which service flags available on this arc	set

4.5 PTL Linehaul Arc Parameters

For each $(i, j) \in A_{\text{LH-PTL}}$:

Parameter	Symbol	Description	Unit
Carrier	$\text{car}(i, j)$	PTL specialist (typically a different carrier from LTL or FTL)	—
Base rate	r_{ij}^{PTL}	Per-shipment + per-mile or pallet-based	USD
Min charge	$\min_{ij}^{\text{PTL}}$	Minimum per-shipment charge	USD
Max payload	$W_{ij}^{\text{PTL}, \text{max}}$	PTL upper weight limit ($\sim 25,000$ kg typ.)	kg
Min payload	$W_{ij}^{\text{PTL}, \text{min}}$	PTL lower threshold ($\sim 4,500$ kg or ~ 6 pallets)	kg
Transit time	μ_{ij}^{PTL}	1–3 days direct	days
Tender accept prob	p_{ij}^{acc}	As FTL	—
Re-tender premium	ρ_{ij}^{re}	As FTL	—

4.6 LTL Linehaul Arc Parameters

For each $(i, j) \in A_{\text{LH-LTL}}$ where $i \in N_{\text{SC}, \text{O}}$, $j \in N_{\text{SC}, \text{D}}$:

Parameter	Symbol	Description	Unit
Carrier	$\text{car}(i, j)$	LTL carrier ID	—
Pickup cutoff	$\text{CGC}_{ij}(\tau)$	Pickup cutoff for day τ (hours-of-day collapsed to day cutoff)	day
Transit time	μ_{ij}^{LTL}	Carrier-published days, origin SC to dest SC (embeds internal hub-and-spoke)	days
Rate table	$R_{ij}(\text{class}, b)$	USD/CWT for each (class, weight-break b) combination	USD/CWT
Min charge	$\min_{ij}^{\text{LTL}}$	Minimum per-shipment charge	USD
Absolute min charge	AMC_{ij}	Higher floor for specific cases (long-haul, oversized)	USD
Fuel surcharge	fsc_{ij}	Pct of linehaul, weekly refresh	—
Max linear feet	$L_{ij}^{\text{LTL}, \max}$	Capacity-load trigger; default 12 ft	ft
Max piece length	$L_{ij}^{\text{LTL}, \text{piece}, \max}$	Single-piece length refusal threshold (carrier-specific, 8–16 ft)	ft
Max piece weight	$W_{ij}^{\text{LTL}, \text{piece}, \max}$	Single-piece weight refusal threshold ($\sim 2,500$ lb)	kg
Max shipment weight	$W_{ij}^{\text{LTL}, \max}$	Total shipment refusal threshold ($\sim 20,000$ – $25,000$ lb)	kg
Tender accept prob	p_{ij}^{acc}	High for LTL (contract): $\sim 0.95+$	—
Re-tender premium	ρ_{ij}^{re}	Modest for LTL (1.05–1.10)	—
Service flags supported	flagsAvail_{ij}	Per-arc service availability flags	set

LTL Weight Break Tiers (NMFTA standard).

Tier code	Breakpoint b	Weight range (lbs)
L5C	0	0–499
M5C	500	500–999
M1M	1,000	1,000–1,999
M2M	2,000	2,000–4,999
M5M	5,000	5,000–9,999
M10M	10,000	10,000–19,999
M20M	20,000	20,000–29,999
M30M	30,000	30,000–39,999
M40M	40,000	40,000+

\$/CWT decreases monotonically through tiers (typical 7.79–9.20% drop per tier; cumulative $\sim 38.66\%$ from L5C to M10M).

4.7 Accessorial & Surcharge Parameters

Charge	Description	Typical
Liftgate	Tail-lift required at pickup or delivery	\$75–150
Residential delivery	Non-commercial address	\$75–100
Inside delivery	Beyond loading dock	\$100–300
Limited access	Schools, military, construction sites	\$75–150
Reconsignment	Destination changed in transit	\$85–200
Notification	Appointment phone-call	\$25
Detention	Driver waiting beyond free time (typ. 2h)	\$50–75/h
Layover	Driver overnight delay	\$250–450/day
Stop-off	Extra pickup or delivery stop	\$50–150
TONU	Truck Ordered Not Used (cancelled pickup)	\$150–250
Chassis day rate	After free time at consignee (drayage)	\$25–40/day
Per diem	Ocean container free time exceeded at consignee	\$75–150/day
Demurrage	Ocean container free time exceeded at terminal	\$100–300/day
Pier pass	LA/LB only	\$35–75

4.8 Tender Acceptance Cost Adjustment

Assumption. Tender acceptance is modeled as a deterministic per-arc cost adjustment in MVP. Given a base rate, the optimizer sees an *expected cost* that captures both the probability of acceptance at the quoted rate and the markup if re-tendered after rejection. The probabilistic model is:

$$c_{ij}^{\text{exp}}(k) = p_{ij}^{\text{acc}} \cdot c_{ij}^{\text{base}}(k) + (1 - p_{ij}^{\text{acc}}) \cdot \rho_{ij}^{\text{re}} \cdot c_{ij}^{\text{base}}(k) \quad (2)$$

Equivalently:

$$c_{ij}^{\text{exp}}(k) = c_{ij}^{\text{base}}(k) \cdot [1 + (1 - p_{ij}^{\text{acc}}) (\rho_{ij}^{\text{re}} - 1)] \quad (3)$$

Interpretation: If the tender is accepted at the quoted base rate (prob p_{ij}^{acc}), the forwarder pays c_{ij}^{base} . If rejected (prob $1 - p_{ij}^{\text{acc}}$), the forwarder re-tenders at a premium ρ_{ij}^{re} on the base rate (typical 1.15–1.25 for FTL spot, 1.05–1.10 for LTL contract). The expected cost combines these.

Worked example. FTL spot lane with $p_{ij}^{\text{acc}} = 0.60$, $\rho_{ij}^{\text{re}} = 1.20$, $c_{ij}^{\text{base}} = \$1,500$:

$$c_{ij}^{\text{exp}} = 1500 \cdot [1 + 0.40 \cdot 0.20] = 1500 \cdot 1.08 = \$1,620$$

The optimizer compares routes using c_{ij}^{exp} ; the operations team sees both the nominal quote c_{ij}^{base} and the expected outcome c_{ij}^{exp} in the recommendation surface. Without this adjustment, the system systematically under-quotes by 15–25% relative to actuals — this was identified as the most critical correction in the adversarial critique (§9).

Deferred to P1: explicit modeling of the re-tender process as a stochastic event with re-routing, multi-attempt cost accumulation, and integration with rolling-horizon re-planning. The MVP deterministic adjustment is a first-order approximation.

4.9 Pre-Computed LTL Cost Per Shipment Per Arc

For each shipment k and LTL arc $(i, j) \in A_{\text{LH-LTL}}^k$, the cost is pre-computed:

$$c_{ij}^{\text{LTL,base}}(k) = \max\left(\max(\text{AMC}_{ij}, \min_{ij}^{\text{LTL}}), \min_{b \in B} \frac{\max(w_k^{\text{lb}}, b) \cdot R_{ij}(\text{class}_k^*, b)}{100}\right) \cdot (1 + \text{fsc}_{ij}) \quad (4)$$

where:

- $w_k^{\text{lb}} = w_k \cdot 2.205$ (kg $\rightarrow$ lb conversion)
- $\text{class}_k^* = \text{class}_k^{\text{FAK}}$ if applicable, else $\text{class}_k^{\text{NMFC}}$
- The inner min over tiers implements the **deficit weight rule**: pay at the lowest-cost tier even if it requires bumping chargeable weight to that tier’s breakpoint
- The outer max applies the minimum charge floor and absolute minimum charge (whichever is higher)

The expected cost after tender adjustment:

$$c_{ij}^{\text{LTL,exp}}(k) = c_{ij}^{\text{LTL,base}}(k) \cdot [1 + (1 - p_{ij}^{\text{acc}})(\rho_{ij}^{\text{re}} - 1)] \quad (5)$$

Both $c_{ij}^{\text{LTL,base}}$ and $c_{ij}^{\text{LTL,exp}}$ are parameters in the MILP — not runtime decision variables, since w_k and class_k are inputs.

FAK class semantics. Shipper-carrier contracts often collapse multiple NMFC classes into a single FAK class for billing simplicity. Example: ”FAK 92.5 for classes 60–200, FAK 175 for 250–400.” For shipment k under such an agreement, $\text{class}_k^{\text{FAK}}$ overrides $\text{class}_k^{\text{NMFC}}$ in the rate lookup. Without FAK support, the model would mis-quote LTL costs for the majority of mid-market shippers.

5 Decision Variables

Variable	Type	Description	Domain
x_{ij}^k	Binary	1 if shipment k uses arc $(i, j) \in A^k$	$\{0, 1\}$
$y_{k,c,ij}$	Binary	1 if shipment k loaded in FTL/PTL truck slot c on linehaul arc (i, j)	$\{0, 1\}$
$z_{c,ij,e}$	Binary	1 if FTL truck slot c on arc (i, j) uses equipment type e	$\{0, 1\}$
$t_k(i)$	Integer	Day-bucket arrival time of shipment k at node i	$\mathbb{Z}_{\geq 0}$

Slot indexing and bound. For each FTL/PTL linehaul arc (i, j) , slots $c \in C_{ij}$ with upper bound:

$$|C_{ij}| = \left\lceil \frac{\sum_{k:(i,j) \in A_{\text{LH-FTL}}^k \cup A_{\text{LH-PTL}}^k} v_k}{\eta \cdot V_{\min}} \right\rceil$$

where $V_{\min}$ is the smallest equipment volume (48’). Symmetry breaking applied (see §6).

Time discretization. $t_k(i)$ is integer (day bucket), not continuous. Discretizing to days reduces big-M constraint counts and yields 10–100 $\times$ solve-time improvement at no realism loss (LTL and FTL transit times are quoted in days, not hours).

6 Constraints

P.1 — Flow Conservation

For each k and $n \in N_k$:

$$\sum_{j:(n,j) \in A^k} x_{nj}^k - \sum_{i:(i,n) \in A^k} x_{in}^k = \begin{cases} +1 & n = o(k) \\ -1 & n = d(k) \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

P.2 — FTL/PTL Truck Volume Capacity

For each linehaul arc $(i, j) \in A_{\text{LH-FTL}} \cup A_{\text{LH-PTL}}$ and slot $c \in C_{ij}$:

$$\sum_{k \in K} v_k \cdot y_{k,c,ij} \leq \eta \cdot \sum_{e \in \mathcal{E}} V_e \cdot z_{c,ij,e} \quad (7)$$

P.3 — FTL/PTL Truck Weight Capacity

$$\sum_{k \in K} w_k \cdot y_{k,c,ij} \leq \sum_{e \in \mathcal{E}} W_e \cdot z_{c,ij,e} \quad (8)$$

P.4 — Single Equipment Type per Slot

$$\sum_{e \in \mathcal{E}} z_{c,ij,e} \leq 1 \quad (9)$$

P.5 — Arc-to-Slot Linkage

For each shipment k and FTL/PTL linehaul arc $(i, j) \in A_{\text{LH-FTL}}^k \cup A_{\text{LH-PTL}}^k$:

$$x_{ij}^k = \sum_{c \in C_{ij}} y_{k,c,ij} \quad (10)$$

P.6 — LTL Linear-Foot Rule (Hard Refusal)

For each shipment k and LTL arc $(i, j) \in A_{\text{LH-LTL}}$:

$$\ell_k \cdot x_{ij}^k \leq L_{ij}^{\text{LTL,max}} \quad (11)$$

If $\ell_k > L_{ij}^{\text{LTL,max}}$ (typically 12 ft), the shipment cannot use this LTL arc. **Pre-processing exclusion:** simply remove (i, j) from A^k for such shipments.

P.7 — LTL Piece Dimension and Weight Refusal

For each LTL arc (i, j) and shipment k with $L_k^{\text{max}} > L_{ij}^{\text{LTL,piece,max}}$, or $W_k^{\text{max}} > W_{ij}^{\text{LTL,piece,max}}$, or $w_k > W_{ij}^{\text{LTL,max}}$:

$$x_{ij}^k = 0 \quad (12)$$

Pre-processing exclusion: arc removed from A^k .

P.8 — LTL Pickup Cutoff

For each shipment k and LTL pickup-related arc:

$$t_k(i) \leq \text{CGC}_{ij}(\tau^*) + M(1 - x_{ij}^k) \quad (13)$$

where τ^* is the day the pickup is scheduled.

P.9 — Time Propagation (day-bucket)

For each arc $(i, j) \in A^k$:

$$t_k(j) \geq t_k(i) + \mu_{ij} - M(1 - x_{ij}^k) \quad (14)$$

μ_{ij} is the arc transit time in days.

P.10 — Delivery Window / MABD

$$T_k^{\text{open}} \leq t_k(d(k)) \leq T_k^{\text{close}} \quad (15)$$

Hard window. Setting $T_k^{\text{open}} = 0$ relaxes the lower bound; setting $T_k^{\text{close}} = T_{\text{max}}$ relaxes the upper bound. Mid-market forwarder MVP defaults to wide windows unless customer specifies appointment.

P.11 — Carrier-Lane Service Availability

For each shipment k and arc (i, j) :

$$x_{ij}^k = 0 \quad \text{if } \text{flags}_k \not\subseteq \text{flagsAvail}_{ij} \quad (16)$$

If shipment requires liftgate but the carrier doesn't offer liftgate on this lane: arc excluded. Pre-processing.

P.12 — Contract FTL Allocation Cap

For each contract FTL lane (i, j) with $\text{ct}(i, j) = \text{contract}$, and period τ :

$$\sum_{c \in C_{ij}} \sum_{e \in \mathcal{E}} z_{c,ij,e} \cdot \mathbb{I}[\text{week}(c) = \tau] \leq \text{alloc}_{ij}(\tau) \quad (17)$$

Forwarder cannot exceed contracted truck allocation per week on contract lanes. Parallels Ocean FCL string allocation (P.3 in ocean model).

P.13 — Cargo Type Compatibility

For shipments requiring DGR / temp-controlled / oversize handling, restrict arc set to compatible carriers (pre-processing).

P.14 — Equipment Compatibility (Container-to-Chassis)

For drayage arcs where shipment k is an FCL container (20' TEU or 40' FEU):

- TEU shipment requires a 20' chassis truck (excluded from 40'/53' chassis arcs)
- FEU shipment requires a 40' chassis truck (excluded from 20'/53' chassis arcs)
- One container per chassis per truck (hard).

P.15 — Symmetry Breaking on Slot Index

For each linehaul arc (i, j) and equipment e :

$$z_{c,ij,e} \leq z_{c-1,ij,e} \quad \forall c \geq 2 \quad (18)$$

Forces filling slot 1 of equipment type e before slot 2.

Additionally, on shipment-to-slot assignment, impose lex ordering: shipment with smallest k index uses lowest-numbered slot when feasible. See §8 for the implementation discussion.

P.16 — Budget Cap (optional)

$$\text{cost}_k \leq B_k \quad \forall k : B_k < \infty \quad (19)$$

P.17 — Domain

$$x_{ij}^k, y_{k,c,ij}, z_{c,ij,e} \in \{0, 1\}; \quad t_k(i) \in \mathbb{Z}_{\geq 0} \quad (20)$$

7 Objective Function

Minimize total expected cost:

$$\begin{aligned} \min \quad & \sum_{k \in K} \sum_{(i,j) \in A_{\text{PU}}^k \cup A_{\text{FD}}^k} c_{ij}^{\text{G}} \cdot x_{ij}^k && (\text{pickup} + \text{delivery ground}) \\ & + \sum_{(i,j) \in A_{\text{LH-FTL}}^k} \sum_{c \in C_{ij}^k} \sum_{e \in \mathcal{E}} c_{ij,e}^{\text{FTL,exp}} \cdot z_{c,ij,e} && (\text{FTL truck cost (per-truck, expected)}) \\ & + \sum_{k \in K} \sum_{(i,j) \in A_{\text{LH-PTL}}^k} c_{ij}^{\text{PTL,exp}}(k) \cdot x_{ij}^k && (\text{PTL per-shipment cost (expected)}) \\ & + \sum_{k \in K} \sum_{(i,j) \in A_{\text{LH-LTL}}^k} c_{ij}^{\text{LTL,exp}}(k) \cdot x_{ij}^k && (\text{LTL pre-computed cost (expected)}) \\ & + \sum_{k \in K} \text{accessorials}(k) + \sum_{k \in K} \text{drayage_holding}(k) && (\text{accessorials} + \text{chassis/per diem/demurrage}) \end{aligned}$$

The FTL cost $c_{ij,e}^{\text{FTL,exp}}$ is per-truck per-equipment-type, computed via Eq. (2):

$$c_{ij,e}^{\text{FTL,exp}} = (r_{ij}^{\text{FTL}} \cdot \text{dist}_{ij})(1 + \text{fsc}_{ij}) \cdot [1 + (1 - p_{ij}^{\text{acc}})(\rho_{ij}^{\text{re}} - 1)]$$

8 Symmetry Breaking — Implementation Notes

What is symmetry in MILP? The slot index c on FTL/PTL trucks is symbolic — “slot 1” and “slot 2” on the same lane with the same equipment are interchangeable from the optimizer’s perspective. If a solution assigns shipment k_a to slot 1 and shipment k_b to slot 2, then the assignment $k_a \rightarrow \text{slot 2}, k_b \rightarrow \text{slot 1}$ is an *equivalent solution* with identical cost. The LP relaxation cannot distinguish these, and the branch-and-bound tree explores both, wasting solve time.

Why it matters at scale. With $|C_{ij}| = 10$ slots per lane and $|K_{ij}| = 30$ shipments routable on the lane, the number of equivalent assignments is $|C_{ij}|! / (|C_{ij}| - |K_{ij}|)!$ for any feasible assignment. Without symmetry breaking, solver time grows multiplicatively.

Lex-order constraints (P.15) — explanation in plain code terms. The constraint $z_{c,ij,e} \leq z_{c-1,ij,e}$ encodes: *slot c can be used only if slot $c - 1$ is also used*. In Python-equivalent pseudocode the optimizer sees:

```
# For each lane (i,j) and equipment type e:
for c in range(2, max_slots + 1):
    model.add_constraint(z[c, ij, e] <= z[c - 1, ij, e])
```

Effect: only solutions where slots are filled in increasing order are feasible. The solver no longer explores “slot 5 used but slot 3 empty” — it always packs into the lowest available slot first. Solver time typically drops 5–20× on instances with significant slot symmetry.

Additional symmetry-breaking on shipment assignment (optional in MVP). A secondary symmetry exists in $y_{k,c,ij}$: two interchangeable shipments could swap slot assignments. This is harder to break without losing feasibility. MVP relies on slot lex-ordering only; full assignment lex-ordering is deferred.

Implementation guidance in code.

- Compute $|C_{ij}|$ before adding y and z variables. Don’t over-allocate — excessive slot counts widen the search space even with lex constraints.
- Add P.15 in the model-build step right after z variable creation.
- Use a solver with strong branch-and-bound (HiGHS, Gurobi, CBC). HiGHS auto-detects some symmetry; lex-ordering helps it further.
- Profile solve time: with and without lex-ordering on a representative 50-shipment instance. Document the speedup. Symmetric-tight instances (many similar shipments on a single lane) benefit most.

9 Critique Findings Incorporated

An adversarial review of the initial design (SESSION_LOG.md, Session 9) identified the following corrections, all incorporated in this draft:

1. **Partial Truckload (PTL) added as third mode.** Original design had only FTL and LTL. The 10,000–25,000 kg / 6–18 pallet band is the dominant forwarder economic regime and is neither traditional FTL nor LTL.
2. **(Carrier, origin SC, destination SC) tuples replace intermediate hub nodes.** Powell & Sheffi load planning is the *carrier’s* internal problem; as forwarders we tender to a carrier and the carrier routes through their own hubs.
3. **LTL hard refusal constraints (P.6, P.7).** Linear-foot rule, piece-length cap, piece-weight cap, total-shipment-weight ceiling. Real-world refusals, not soft cost trade-offs.
4. **Contract FTL allocation cap (P.12).** Parallels ocean string allocation; the model without this assumes infinite contract capacity.
5. **Delivery window / MABD (P.10).** Big-box consignees enforce appointment windows with chargebacks. Modeled as $[T_k^{\text{open}}, T_k^{\text{close}}]$ window; default to wide window if no appointment.

6. **Carrier-lane service availability matrix (P.11).** Liftgate, residential, limited access, inside delivery as feasibility flags, not just cost.
7. **FAK class override.** Shipper-contract-specific class collapse, not global NMFC lookup.
8. **Tender acceptance probability (§4.8).** Deterministic per-arc cost adjustment incorporating p^{acc} and ρ^{re} . Critical for accurate quotes; deterministic-only cost would under-quote by 15–25%.
9. **Chassis day rate, per diem, demurrage, pier pass** as drayage cost components.
10. **Time discretization to days; lex-order on slot index c .** Tractability fixes; 10–100× solve-time win.

10 Open Items and Future Extensions

The following are explicitly deferred to P1 and beyond:

1. **Stochastic tender acceptance modeling.** MVP uses deterministic expected-cost adjustment. P1: explicit Monte Carlo over tender attempts; re-routing on rejection via rolling-horizon controller.
2. **Driver Hours of Service (DOT).** 11 h driving, 14 h on-duty, 70-in-8 cycle. Affects which lanes a single-driver truck can serve in one shift; team-driver operations relaxes. MVP assumes all lanes are operationally feasible.
3. **Backhaul / round-trip optimization.** Optimizing the truck’s return load. Carrier-side problem; relevant for dedicated lane pricing.
4. **Driver / specific-truck dispatch.** Which driver, which truck, which load. Carrier-side problem.
5. **Specialty equipment.** Reefer, flatbed, lowboy, hazmat-certified, oversize permits, intermodal-only chassis types.
6. **Dedicated lanes / weekly milk runs.** Recurring scheduled FTL with fixed capacity. Important for high-volume forwarders.
7. **Pool distribution.** LCL/PTL freight broken down at an intermediate pool point and re-distributed via local LTL or last-mile.
8. **Cross-border (Mexico, Canada) trucking.** US-MX customs at land border; bonded carriers; B1 visa drivers; transflo handoffs.
9. **Stop-off charges, TONU, reweigh / reclass variance.** Modeled as flat accessorial in MVP; full operational treatment via agent layer feedback.
10. **Spot rate stochastic availability.** Spot capacity not always available on the requested day; subsumes under tender acceptance probability for now.
11. **Twin-20 combo chassis at port.** In-port two-TEU drayage move. Rare; deferred.
12. **Drop-and-hook operations.** Carrier drops a pre-loaded trailer at shipper and picks up a loaded one in return. Affects scheduling but not routing.
13. **Dock appointment scheduling.** MVP treats appointments as windows; P1 could query consignee appointment availability via API.
14. **LTL re-classification variance.** Reweigh/reclass at carrier dock can change post-billed cost from quoted cost by 5–15%. Flag to agent layer.

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